LESSON DETAILS

Resource: Geometric Transformations Ugly Holiday Sweater Project

Suggested Class Time: 30-45 minutes

The Big Idea

Students will be able to apply what they know about translations, reflections and rotations to create an "Ugly Holiday Sweater". The project includes one self-checking activity and one free-form activity.

Standards

CCSS	TEKS
• 8.G.A.2: Understand that a two-dimensional figure is congruent to another if the second can be obtained from the first by a sequence of rotations, reflections, and translations; given two congruent figures, describe a sequence that exhibits the congruence between them. • 8.G.A.3: Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates. • 8.G.A.4: Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations; given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them.	• 8.3(B): Compare and contrast the attributes of a shape and its dilation(s) on a coordinate plane. • 8.3(C): Use an algebraic representation to explain the effect of a given positive rational scale factor applied to two-dimensional figures on a coordinate plane with the origin as the center of dilation.

Horizontal Planning

Pre-requisite Skills	Current Skills	Upcoming Skills
• Basic shape transformations	• Geometric Transformations	• Pythagorean Theorem

Common Misconceptions

- Students can mix up clockwise and counter-clockwise
- Students learn in elementary school that these relationships are called "flips, turns, and slides". They may have difficulty with these new titles.
- Rotations involve degree measurements. If students are not familiar with common rotations (like 90o or 180o), it will be difficult to complete these activities.

Materials Needed

- One copy of the directions per student
- Two copies of the coordinate plane – one for each activity – per student
- Two copies of the sweater – one for each activity – per student
- Crayons, markers, and/or colored pencils

Directions

- **One:** Students will be given one copy of the directions, one copy of the coordinate graph and one copy.
- **Two:** Students will use the coordinate graph as a "rough draft" to figure out where the sweater decorations will go. They will follow through the directions and come up with the placement of each object. They'll have the chance to add to each object when they are finished – for example, they will turn the rectangle into a present.
- **Three:** They will then use the sweater with the grid inlay to sketch out their ugly sweater. You can have them add additional decorations / colors / designs / patterns to the sleeves if you wish.
- **Four:** The second part of the project allows the students to come up with their own ugly sweater design. They must write their own directions (using translations, reflections and/or rotations – your choice – and can then switch with a classmate to make additional sweaters.)

Credits :

- Children Clipart from Melonheadz Illustration
- Blank Sweater Template from Glitter Meets Glue Designs

The Ugly Christmas Sweater Math Project

directions:

1. The first sheet you will use will be the one without the sweater. This will be your "rough draft".
2. Perform the indicated transformations on each of the five shapes. You can use a blank coordinate plane or piece of graph paper if you need more space. Mark off the location of each shape after the transformations.
3. Copy the final shapes from the rough draft onto the coordinate plane on the "Ugly Sweater" coordinate plane. Pay careful attention to the coordinates of each vertex as you copy each shape.
4. Once all of the shapes are on the sweater, you will "decorate" each shape to turn it into a holiday object.
   - Shape 1 : Decorate as a Christmas tree
   - Shape 2 : Decorate as an ornament
   - Shape 3 : Decorate as a present
   - Shape 4 : Decorate as an ornament
   - Shape 5 : Decorate the star to make it festive
   - Shape 6 : Decorate as another present

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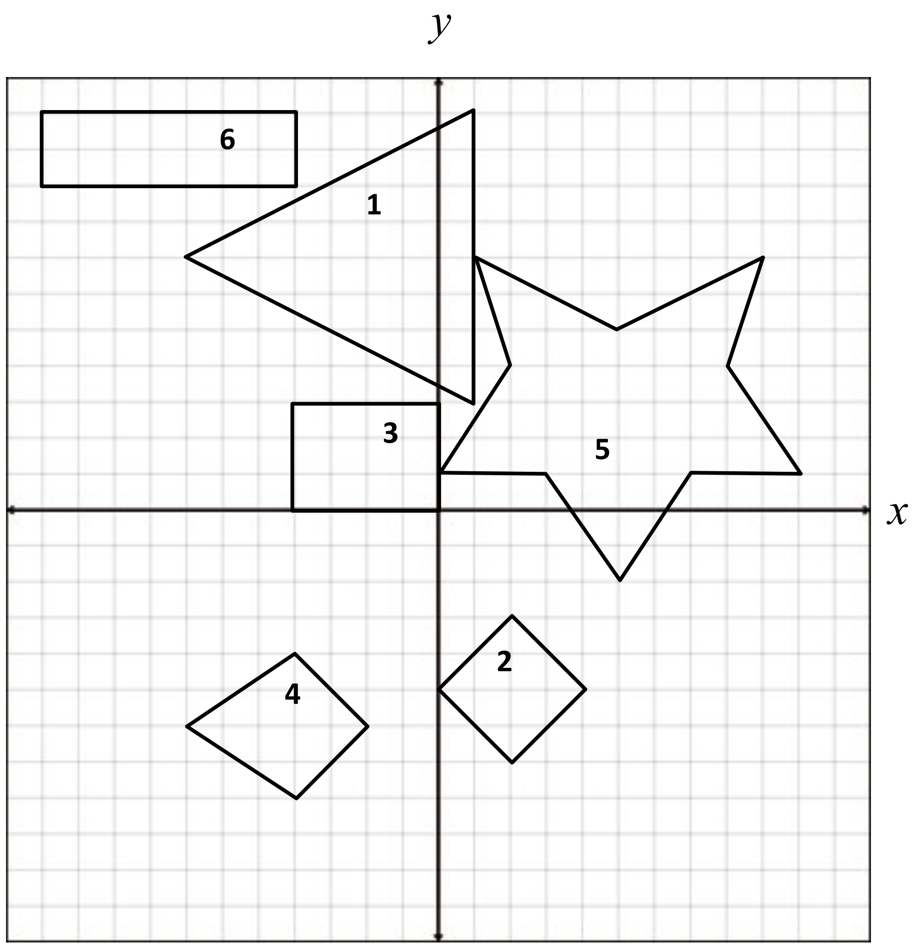
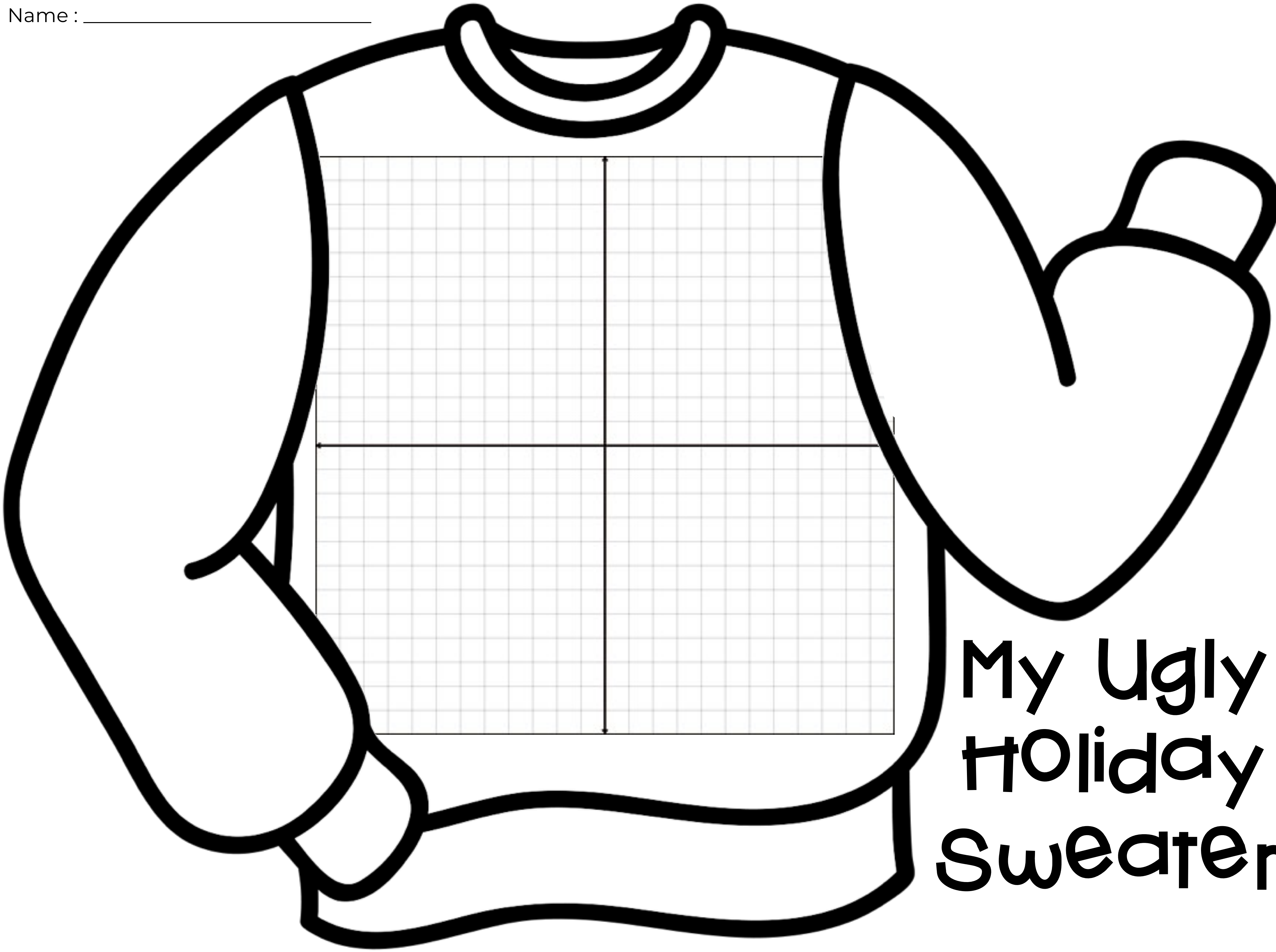
The Ugly Holiday Sweater
math project



Perform the transformations on each shape:

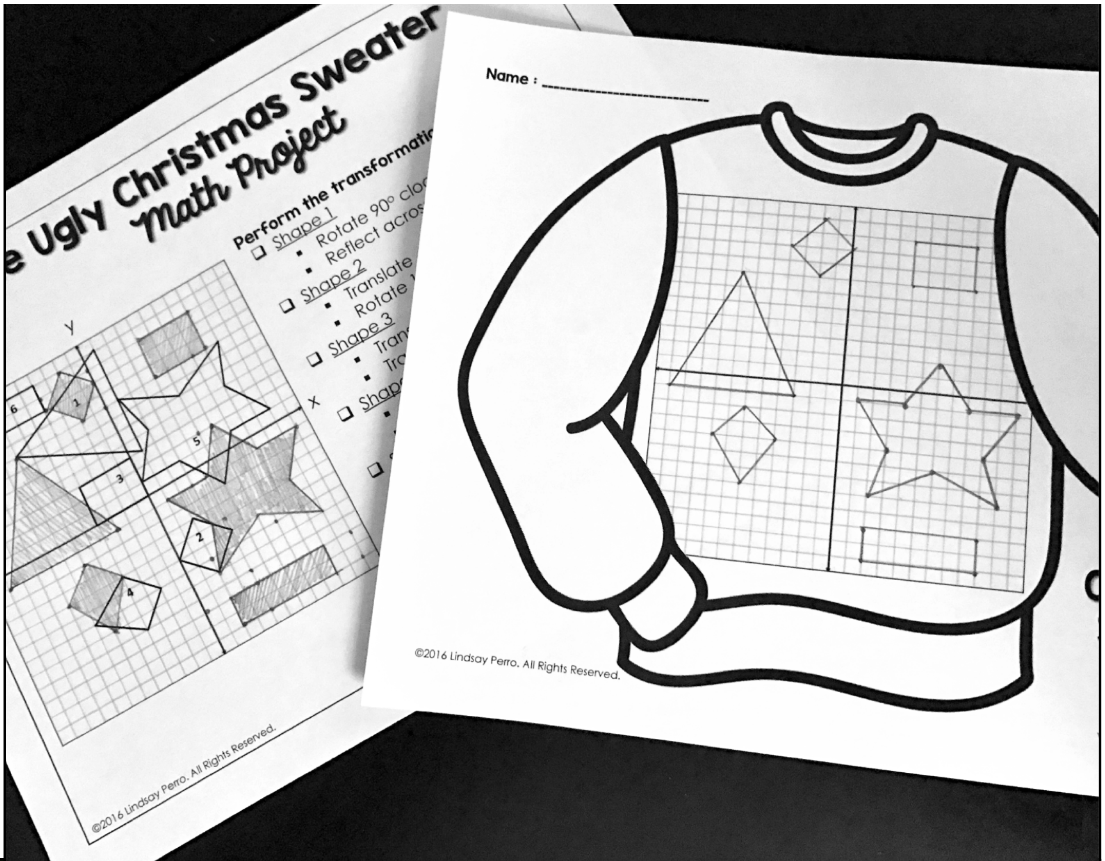
- ❑ Shape 1
  - Rotate 90° clockwise
  - Reflect across the y-axis
- ❑ Shape 2
  - Translate 4 units down
  - Rotate 180°
- ❑ Shape 3
  - Translate 7 units up
  - Translate 8 units right
- ❑ Shape 4
  - Reflect across the x-axis
  - Rotate 90° counterclockwise
- ❑ Shape 5
  - Reflect across the x-axis
  - Translate 1 unit right
- ❑ Shape 6
  - Rotate 180°
  - Translate 2 units left

Name : ____________________



My Ugly Holiday Sweater

ANSWER KEY


Ugly Christmas Sweater
Math Project
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– Lindsay

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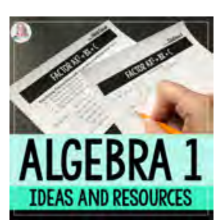

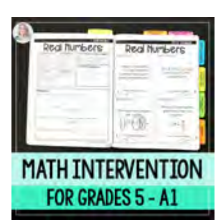

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